Geometric motivic integration part 1.

Now: Basics needed to define motivic integrals.
using arc-space $\mathcal{L}(X)$ of $\mathbb{C}[[t]]$ pts.

- Nice table illustrating analogy between p-adic and motivic measure

<u>Def. 2.1.</u> R Ring, $\lambda : Obj(Var_{\mathbb{C}}) \to R$ with

- $\lambda(X) = \lambda(Y)$ if $X \cong Y$
- $\lambda(X) = \lambda(X \setminus F) + \lambda(F)$ for $F \subset X$ zar closed.
- $\lambda(X \times Y) = \lambda(X) \cdot \lambda(Y)$

is an additive invariant of complex varieties.

<u>Examples</u>: (1) Euler char $\chi(X) = \sum_{i \geq 0} (-1)^i \dim H^i(X, \mathbb{Q})$.
Add inv for cplx var. In general need to use $\chi_c(X)$.

(2) To smooth X associate Virtual Hodge-Deligne Polynomial:

$$H_X(u,v) = \sum_{p,q \geq 0} (-1)^{p+q} h^{p,q}(X) u^p v^q \in \mathbb{Z}[u,v]$$

or virtual Poincare $H_X(t,t)$. Note $H_X(1,1) = \chi(X)$.

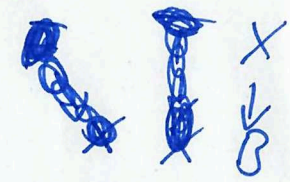
(3) Counting points for var / $\mathbb{Q}$.

<u>Def</u>: Grothendieck ring of vars $(K_0(Var_{\mathbb{C}}), +, \cdot)$ is gen by $[X]$ with relations $[X] = [Y]$ if $X \cong Y$
$[X] = [X \setminus F] + [F]$ $F \subset X$ zar-closed.
$[X \times Y] = [X] \cdot [Y]$

$\mathbb{L} = [\mathbb{A}^1_{\mathbb{C}}]$ is the Lefschetz motive.

<u>Examples</u>: - $[\mathbb{C}^n] = \mathbb{L}^n$ $\quad [\mathbb{C}^*] = \mathbb{L} - 1$

- $[\mathbb{C}P^n] = \sum_{i=0}^{n} [\mathbb{C}]^i$
- $[P_m] = [P_m \setminus 0] = m \cdot [\mathbb{C}^*] + [0] = m \cdot \mathbb{L} - (m-1)$
- $[\{y^2 - x^3\}] = [\mathbb{C}^*] + pt = \mathbb{L}$
- Can also define $[C]$ for contractible $C = \sqcup U_i$ as $\sum [U_i]$.
- Product extends to locally trivial fibrations with fibre F: $[B] \cdot [F] = [X]$.



Have seen that we can generate $K_0(\mathrm{Var}_{\mathbb{C}})$ using smooth $[X]$.
Actually get it from introducing the following relation for smooth proj. $Y \subset X$:

$$[Bl_Y[X]] - [E] = [X] - [Y].$$

<u>Thm. 2.8.</u> (Universality)

Any additive inv of cplx varieties factors through $K_0(\mathrm{Var}_{\mathbb{C}})$!

$\rightsquigarrow$ Have ring morphisms

$$\begin{tikzcd} K_0(\mathrm{Var}_{\mathbb{C}}) \arrow[r, "H"] \arrow[rr, bend right, "N^p"'] & \mathbb{Z}[u,v] \arrow[r, "u=v"] \arrow[u, bend left, "\chi"] \arrow[r, bend right, "p"'] & \mathbb{Z}[t] \arrow[d] \\ & & \mathbb{Z}. \end{tikzcd}$$

- Idea: $[X]$ is most general way to "count pts" or "measure size".

$$p = |\mathbb{A}^1_{\mathbb{F}_p}| \rightsquigarrow \mathbb{L} = |\mathbb{A}^1_{\mathbb{C}}|$$

- Can see $\mathbb{Z}[\mathbb{L}] \subsetneq K_0(\mathrm{Var}_{\mathbb{C}})$ as $H(\mathbb{L}^i) = (uv)^i$ but genus g curve has $H(C) = 1 - gu - gv + uv$.

It is complicated: Has zero-divisors; $\mathbb{L}$ is one; but not nilpotent. ($\chi(\mathbb{L})=1$). 3

If $[X]=[Y]$, X and Y admit decomp into iso subvars: True in dim 1, false in general.

2.2. Jets and arcs.

Def $X = V(f_1 = \dots = f_r = 0)$ in $\mathbb{C}^d$

- m-jets of X: $\mathcal{L}_m(X)$ is alg var over $\mathbb{C}[t]/t^{m+1}$

$$\mathcal{L}_m(X) = \{ \gamma_m = (x_1(t), \dots, x_d(t)) \in (\mathbb{C}[t]/t^{m+1})^d,\ f_i(\gamma_m) = 0 \}.$$

- formal arc-space $\mathcal{L}(X)$: now are $\mathbb{C}[[t]]$

For m-jet or arc say: $\gamma_m(0)$ or $\gamma(0)$ is origin on X.

Rmk 2.12. Have maps $(m \geq n)$ $\pi_n^m : \mathcal{L}_m(X) \to \mathcal{L}_n(X)$ $\pi_n : \mathcal{L}(X) \to \mathcal{L}_n(X)$.

which compose as expected.

Ex: $\mathcal{L}_m(\mathbb{C}^d) = \mathbb{C}^{d(m+1)}$; Also clear $\mathcal{L}_m(X) \subseteq \mathbb{C}^{d(m+1)}$ and maps π_n^m are just proj onto $d(n+1)$ coord.

$X = \{y^2 - x^3\}$

- $\mathcal{L}_0(X) = X$

$\mathcal{L}_1(X) = \{ (a_0 + a_1 t, b_0 + b_1 t) \mid (b_0 + b_1 t)^2 - (a_0 + a_1 t)^3 = 0 \mod t^2 \}$.

$b_0^2 - a_0^3 = 0 \quad 2b_0 b_1 - 3a_0^2 a_1 = 0.$

- Fiber at $(0,0)$ is whole (a_1,b_1) plane; which is $T_{(0,0)}X \simeq \mathbb{C}^2$
- $(a,b) \neq (0,0)$ fiber is a line

In the same way: $\mathcal{L}_2(X) \subseteq \mathbb{C}^6$ $\begin{cases} b_0^2 - a_0^3 = 0 \\ 2b_0b_1 - 3a_0^2 a_1 = 0 \\ b_1^2 + 2b_0b_2 - (3a_0a_1^2 + 3a_0^2 a_1) = 0 \end{cases}$

Fiber of π_0^2 at $(0,0)$ is ~~plane~~ 3-fold $\{(0,0,a_1,0,a_2,b_2)\} \simeq \mathbb{C}^3$

but its image by π_1^2 is the line $\{a_0 = b_0 = b_1 = 0\} \subseteq \mathbb{C}^4$.

So $\pi_1^2 : \mathcal{L}_2(X) \to \mathcal{L}_1(X)$ is not surjective

However: π_m^{m+1} is surj. above non-sing part with fiber $\mathbb{C}$.

<u>Rmk</u>: Can do for any variety via

$$\mathcal{L}_m(X) = \mathrm{Hom}(\mathrm{Spec}(\mathbb{C}[t]/t^{m+1}), X).$$

and $\mathcal{L}(X)$ as proj. limit. Functorial.

<u>Prop 2.17.</u>
- $\mathcal{L}_0(X) = X$ $\quad \mathcal{L}_1(X) = TX$
- X smooth: π_n^m is Zar-loc triv. fibration; in part surj. and $\dim \mathcal{L}_m(X) = (m+1)d$.
- For any arc γ: Fiber $\pi_m^{m+1}(\gamma)$ is empty or iso to $T_{x_0}X$.
- Closure of $\pi_0^{m-1}(X_{reg})$ is irred comp of $\mathcal{L}(X)$

2.18. $\mathcal{L}_m(X)$ irred for any $m>0 \iff X$ has rat sing.

Ex. 2.19. $\mathcal{L}_m(X)$ vs. $\pi_m(\mathcal{L}(X))$

Know $\pi_m(\mathcal{L}(X)) \subseteq \mathcal{L}_m(X)$ but not any jet can be lifted to arc.

$X=\{xy=0\}$ $\qquad \mathcal{L}(X) = \{(\underbrace{x(t)}_{\sum a_i t^i}, \underbrace{y(t)}_{\sum b_i t^i}) \mid x(t)y(t)=0\}$

Have cond: $\displaystyle\sum_{i=0}^{k} a_i b_{k-i} = 0 \quad \forall k.$

and $\mathcal{L}_m(X) = \{(a_0,\dots,a_m,b_0,\dots,b_m) \mid \text{is true for } k=0\dots m\}.$

So $\mathcal{L}_1(X) = \{(a_0+a_1t, b_0+b_1t) \mid a_0b_0=0 \quad a_0b_1+a_1b_0=0\}.$

$= \underbrace{\{a_0=a_1=0\}}_{V_{2,0}} \cup \underbrace{\{a_0=b_0=0\}}_{V_{1,1}} \cup \underbrace{\{b_0=b_1=0\}}_{V_{0,2}}$ Three comp $\mathbb{C}^2$.

$\pi_1(\mathcal{L}(X))$? $W_{i,j} := V_{i,j} \cap \pi_1(\mathcal{L}(X))$

- $W_{2,0}$: can lift all of them
  $W_{0,2}$: " " " " "

  $W_{1,1}$: Cond for $k=2$ is $a_1b_1=0 \rightsquigarrow V_{1,1} \subset W_{2,0} \cup W_{0,2}$
  can lift there with $\tilde{\varphi}=0$.

And actually : $\mathcal{L}_m(X) = \bigcup_{l+h=m+1} V_{l,h}$ $\quad V_{l,h} = \{a_0,\dots,a_{l-1}=0, b_0,\dots,b_{h-1}=0\}$ irred comp.

$|\mathcal{L}_m(X)| = (m+2)\mathbb{L}^{m+1} - (m+1)\mathbb{L}^m$

$|\pi_m(\mathcal{L}(X))| = 2\mathbb{L}^{m+1} - 1.$

2.3. Motivic measure

$\mu(\mathcal{L}(\mathbb{O})) = 1$

Want to define a measure μ normalized over $\mathcal{L}(X)$

$$\text{“}\ \mu(C) = \frac{|\pi_m C|}{|(\mathbb{A}^d_k)^{m+1}|} \quad \text{for } m \gg 0 \text{ and}$$

$$C = \left(\pi_m^{-1}(\pi_m(C))\right)$$

“cylinder” $C \subset \mathcal{L}(X)$ (Recall 1.21. For cylinder C

$$\left(\frac{|\pi_m(C)|}{p^{d(m+1)}}\right)_{m \geq 0} \text{ is constant for } m \gg 0.)$$

but not “number of pts” $\leftrightsquigarrow$ class in Grothendieck ring.

- Want $\dfrac{[X]}{\mathbb{L}^n} \rightsquigarrow$ localize at $\mathbb{L}$: $\mathcal{M}_{\mathbb{C}} = K_0(Var_{\mathbb{C}})[\mathbb{L}^{-1}]$.

Any invariant λ with $\lambda(\mathbb{L}) \neq 0$ extends to map

$$\mathcal{M}_{\mathbb{C}} \to R[\lambda(\mathbb{L})^{-1}].$$

In part $\chi, P, H, N^p(\mathbb{L})$
$\quad\quad 1 \quad t^2 \quad w \quad p$

First general cylinder: $A \subset \mathcal{L}(X)$ is called cylinder if $A = \pi_m^{-1}(C_m)$ for some m and some constructible $C_m \subset \mathcal{L}_m(X)$

(1) Morally: set of arcs $A \simeq C_m \times \mathbb{C}^\infty$: Boolean algebra.

(2) If $X = \bigsqcup_{i=0}^k W_i$ then $\mathcal{L}(X) = \bigsqcup_{i=0}^k \pi_0^{-1}(W_i)$.

2.22. X smooth ; $A = \pi_{m_0}^{-1}(C_{m_0})$ a cylinder in $\mathcal{L}(X)$.

Then $\dfrac{|\pi_m(A)|}{\mathbb{L}^{d(m+1)}} \in \mathcal{M}_{\mathbb{C}}$ is constant for any $m \geq m_0$.

proof.: X smooth $\Rightarrow \pi_{m_0}^{m}$ is locally trivial fibration of $\mathcal{L}_{m_0}(X)$ with fiber $\mathbb{C}^{d(m-m_0)}$.

So $|\pi_m(A)| = |C_{m_0}| \cdot |\mathbb{C}^{d(m-m_0)}| = |C_{m_0}| \cdot \mathbb{L}^{(d(m-m_0)}$.

Gen not true :

Def 2.24. $A \subset \mathcal{L}(X)$ stable if for some $m \in \mathbb{Z}_{\geq 0}$ we have:

(1) $\pi_{m_0}(A)$ is contractible and $A = \pi_{m_0}^{-1}(\pi_{m_0}(A))$.

(2) For any $m \geq m_0$ proj. is a piecewise triv fibration with constant fiber $\mathbb{C}^d$. ($\pi_m(A)$ can be partitioned in locally closed S such that any of them admit covering $S = \bigcup U_n$ with $\pi_m^{m+1\,-1}(U_n) \cong U_n \times \mathbb{C}^d$.

Lemma 2.25. $A \subset \mathcal{L}(X)$ cylinder and $A \cap \mathcal{L}(X_{sing}) = \emptyset$, A is stable.

$\rightsquigarrow \lim\limits_{m \to \infty} \dfrac{|\pi_m(A)|}{\mathbb{L}^{d(m+1)}}$ exists in $\mathcal{M}_{\mathbb{C}}$.

$\rightsquigarrow$ obtain additive invariant w.r.t. finite union and intersection

$\tilde{\mu}_{\mathcal{L}(X)} : \{\text{Stable subsets of } \mathcal{L}(X)\} \longrightarrow \mathcal{M}_{\mathbb{C}}$.

"naive motivic measure"

<u>Care:</u> stable subsets do not form algebra, as $\mathcal{L}(X)$ not stable.

For example for $X = \{xy = 0\}$ from ex. 2.19.

we see $$\frac{|\Pi_m(\mathcal{L}(X))|}{\mathbb{L}^{d(m+1)}} = \frac{2\mathbb{L}^{m+1} - 1}{\mathbb{L}^{m+1}} = 2 - \frac{1}{\mathbb{L}^{m+1}}$$ does not stabilize

$\rightsquigarrow$ Extend $\tilde{\mu}_{\mathcal{L}(X)}$ to $\mu_{\mathcal{L}(X)}$

$\|$

2.3.2. Completion of $\mathcal{M}_{\mathbb{C}}$:

complete by a norm in analogy to p-adic : $\mathbb{L}^{-m}$ are small.

<u>Def.</u> $\tau \in K_0(Var_{\mathbb{C}})$ is d-dimensional if

$$\tau = \sum_{i=1}^{j} a_i [X_i] \quad \text{s.t. } d = \max\{\dim X_i\}.$$

extend to $\mathcal{M}_{\mathbb{C}}$ via $\dim \mathbb{L}^{-1} = -1$.

<u>Prop:</u> Virtual dim map is well defined and

(1) $\dim(\tau \cdot \tau') \leq \dim \tau + \dim \tau'$.

(2) $\dim(\tau + \tau') \leq \max\{\dim \tau, \dim \tau'\}$ with "=" if $\dim \tau \neq \dim \tau'$.

Filter $\mathcal{M}_{\mathbb{C}}$ by virtual dim to define completion

$$\mathcal{F}^m \mathcal{M}_{\mathbb{C}} = \{\tau \in \mathcal{M}_{\mathbb{C}} \mid \dim \tau \leq -m\} = \left\langle \frac{[X]}{\mathbb{L}^i} \,\middle|\, \dim X - i \leq -m \right\rangle$$

$$\hat{\mathcal{M}}_{\mathbb{C}} = \varprojlim_m \mathcal{M}_{\mathbb{C}} / \mathcal{F}^m \mathcal{M}_{\mathbb{C}}$$

(1) seq $\frac{[X_n]}{\mathbb{L}^{i_n}} \to 0$ in $\hat{\mathcal{M}}_{\mathbb{C}} \iff \dim X_n - i_n \to -\infty$

(2) $\hat{\mathcal{M}}_{\mathbb{C}}$ is completion w.r.t. norm $\delta : \mathcal{M}_{\mathbb{C}} \to \mathbb{R}_{\geq 0}$

$$\tau \mapsto \delta(\tau) = e^{\dim \tau}$$

is non-arch $\delta(\tau) = 0 \iff \tau = 0$

$$\delta(\tau \cdot \tau') \leq \delta(\tau) \cdot \delta(\tau')$$

If $\mathcal{M}_{\mathbb{C}}$ is not domain then "<".

$$\delta(\tau + \tau') \leq \max\{\delta(\tau), \delta(\tau')\}.$$

Back to $X = \{xy = 0\}$: $\lim_{m \to \infty} \frac{|\pi_m(\mathcal{L}(X))|}{\mathbb{L}^{d(m+1)}} = \lim_{m \to \infty} \left(2 - \frac{1}{\mathbb{L}^{m+1}}\right) = 2.$

<u>Thm</u>: (DL 99) For cylinder $A \subset \mathcal{L}(X)$ $\mu_{\mathcal{L}(X)}(A) = \lim_{m \to \infty} \frac{|\pi_m(A)|}{\mathbb{L}^{d(m+1)}}$

exists and the map

$$\mu_{\mathcal{L}(X)} : \{\text{Cylinder of } \mathcal{L}(X)\} \to \hat{\mathcal{M}}_{\mathbb{C}}$$
$$A \mapsto \mu_{\mathcal{L}(X)}(A)$$

is a σ-additive measure i.e. for p.w. disj. $\{A_i\}$

$$\mu(\sqcup A_n) = \sum_{n \geq 0} \mu_{\mathcal{L}(X)}(A_n).$$

"Motivic measure"

Prop. 2.34 $Z \subseteq X$ proper closed subvar : $\mu_{\mathcal{L}(X)}(\mathcal{L}(Z)) = 0$.

Remark: - Some people normalize $\mu(\mathcal{L}(X)) = [X]$.

- $\mathcal{M}_{\mathbb{C}} \to \hat{\mathcal{M}}_{\mathbb{C}}$ inj? Kennedis $\bigcap_{m \in \mathbb{Z}} \mathcal{F}^m \mathcal{M}_{\mathbb{C}}$.

For ex the Hodge pol factors through it, since

$$\deg H([X_m]\mathbb{L}^{-i_m}) = 2\dim X_m - 2i_m \leq -2m \to -\infty.$$

- Sing case: $\mathcal{L}(X) \setminus \mathcal{L}(X_{sing})$ as union of complements in $\mathcal{L}(X)$ of smaller tub nbhds.

$$\mathcal{L}(X) \setminus \mathcal{L}(X_{sing}) = \bigcup_{e \geq 0} \mathcal{L}^{(e)}(X), \quad \mathcal{L}^{(e)}(X) = \mathcal{L}(X) \setminus \pi_e^{-1}(\mathcal{L}(X_{sing}))$$

- Can do more gen for semi-alg sets.